

Cross-Scale Endpoint-Normalized Radius in a Finite V59 Residual

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Abstract

TPC-269 showed that a registered growing-cutoff proxy and a convex profile path still cross the quarter threshold, but it did not measure the size of the residual radius. We introduce the dimensionless finite observable $X_{N,\theta} = (R_{N,\theta}^2)^3/N^{10}$, equivalently $(R_{N,\theta}/N^{5/3})^6$, so that the endpoint-scale normalization is rational once the upstream interval for R^2 is certified. On the same source-compatible finite registry, four dyadic ratio intervals exhibit the separated pattern DROP–RISE–RISE–DROP: the ratio for $96 \rightarrow 192$ is above 23, while that for $64 \rightarrow 128$ is below $1/4$. Three matched profile controls remain in $(1/2, 3/4)$. This is a finite normalization audit and a scoped stability obstruction; it is not an asymptotic radius estimate, a power-saving theorem, or a twin-prime result.

1 Position and claim firewall

The TPC-266 typed compiler leaves the Schur residual radius as a separate lane. TPC-267 instantiated the literal prime-shell object at finite scale, and TPC-268 showed that declared finite cutoff changes can alter its quarter-sector classification. TPC-269 then used $z_N = \lfloor \log N \rfloor$ and an affine two-profile path, finding a finite profile flip. The remaining minimal question is a scale question: does the radius itself have a stable normalization on the same finite interface?

The status of this paper is

NUMERICALLY_CERTIFIED_FINITE_CROSS_SCALE_RADIUS_NORMALIZATION_AUDIT.

The word “finite” is part of the claim. The registered scales are a diagnostic registry, not an asymptotic sequence with a proved source-level uniformity law. In particular, no observed ratio below or above one is promoted to an eventual lower or upper bound.

2 The frozen finite residual

Let N be one of the registered even scales and write

$$\mathcal{I}_N = \{N/2 + 1, \dots, N\}, \quad \mathcal{Q}_Q = \{q : q \text{ prime}, Q < q \leq 2Q\}.$$

The source and shifted-prime comparison are the finite objects used in TPC-269:

$$\beta_N(t) = \frac{\Lambda(t)}{\log t} - \sum_{d|t, d^{400} \leq N^{133}} \mu(d), \quad w_N(u) = \Lambda(u+2) - b_N^{(z_N)}(u),$$

where $z_N = \lfloor \log N \rfloor$ on the registry and

$$b_N^{(z)}(u) = C_2^{(z)} 1_{2 \nmid u} \prod_{p \leq z} \frac{p}{p-1} \prod_{\substack{p|u \\ p > z}} \frac{p-1}{p-2}.$$

As before, the value is zero when $u+2$ has a prime factor at most z , and the Euler product is enclosed by the finite tail protocol.

For $s = 1, 2$, let

$$K_{H,s}(h) = \left(1 + \frac{h^2}{H^2}\right)^{-s}$$

and retain the prime shell, unit masks, deleted diagonal, and centered residue factor in

$$A_s(u, t) = 1_{u \neq t} \sum_{q \in \mathcal{Q}_Q} q K_{H,s}(u-t) 1_{q \nmid ut} \left(1_{u \equiv t \pmod{q}} - \frac{1}{q-1} \right).$$

For the base rows we use $\theta = 0$, so $g_{N,0} = A_1 \beta_N$. At three matched scales we also use the source-compatible finite mixture

$$A_\theta = (1-\theta)A_1 + \theta A_2, \quad g_{N,\theta} = A_\theta \beta_N, \quad \theta = 1/2.$$

Partition $\mathcal{I}_N$ into four equal blocks and let P_3 be the orthogonal projection onto the three contrasts

$$(1, 1, -1, -1), \quad (1, -1, 0, 0), \quad (0, 0, 1, -1).$$

The residual product is

$$R_{N,\theta}^2 = \|(I - P_3)w_N\|^2 \|(I - P_3)g_{N,\theta}\|^2.$$

All non-logarithmic terms are rational on a finite row. The logarithms and the Euler constant use outward intervals with tail cutoff $P = 50000$.

3 Endpoint normalization

The endpoint budget in TPC-265 uses the baseline exponent $5/3$ for a radius lane. Directly taking $N^{5/3}$ would introduce an unnecessary algebraic number into a finite certificate. We instead define

$$\mathbf{Xi}_{N,\theta} := \frac{(R_{N,\theta}^2)^3}{N^{10}}. \tag{1}$$

Lemma 1 (exact normalization identity). *If $R_{N,\theta}^2 > 0$, then*

$$\mathbf{Xi}_{N,\theta} = \left(\frac{R_{N,\theta}}{N^{5/3}} \right)^6.$$

If $[r_-, r_+]$ is a positive outward interval for R^2 , then

$$\mathbf{Xi} \in \left[\frac{r_-^3}{N^{10}}, \frac{r_+^3}{N^{10}} \right]. \tag{2}$$

Proof. The first identity follows from $R^6 = (R^2)^3$ and $N^{10} = (N^{5/3})^6$, and the second follows because cubing is increasing on the positive half-line. Every endpoint in (2) is rational in the finite interval engine. $\square$

Table 1: Base endpoint-normalized sixth-power intervals.

N	z_N	$\text{Xi}_{N,0}$ interval
64	4	[8.81278791526e-7, 8.81449971308e-7]
96	4	[4.95949095187e-7, 4.96030749164e-7]
128	4	[2.04279432566e-7, 2.04322254881e-7]
192	5	[1.18847779301e-5, 1.18871727315e-5]
256	5	[1.46532173424e-6, 1.46559968348e-6]
384	5	[9.54437330125e-6, 9.54594504625e-6]

For positive intervals $[a_-, a_+]$ and $[b_-, b_+]$, the derived scale ratio uses

$$\frac{[a_-, a_+]}{[b_-, b_+]} = \left[\frac{a_-}{b_+}, \frac{a_+}{b_-} \right]. \quad (3)$$

We study $D(a, b) = \text{Xi}_{b,0}/\text{Xi}_{a,0}$ for the four registered dyadic pairs, and compare $\text{Xi}_{N,1/2}/\text{Xi}_{N,0}$ at three scales.

4 Certified finite result

The six base rows use

$$(N, H, Q) = (64, 15, 4), (96, 20, 5), (128, 24, 5), (192, 32, 6), (256, 38, 6), (384, 50, 7).$$

The resulting $\text{Xi}_{N,0}$ intervals are shown in Table 1; the stored radius-square intervals are inherited from the exact finite operator replay and are not floating-point point estimates.

Theorem 1 (finite cross-scale normalization audit). *On the declared registry, the four dyadic ratios are*

$a \rightarrow b$	$D(a, b)$ interval	finite classification
$64 \rightarrow 128$	$[0.231753859227, 0.231847466257]$	drop below 1/4
$96 \rightarrow 192$	$[23.9597604587, 23.9685339622]$	rise above 16
$128 \rightarrow 256$	$[7.17162080603, 7.17448479796]$	rise above 7
$192 \rightarrow 384$	$[0.802913654645, 0.803207691586]$	drop below 1

Thus the dyadic pattern is

$$\text{DROP} - \text{RISE} - \text{RISE} - \text{DROP}.$$

At the matched scales $N = 96, 128, 256$, the profile ratios obey

N	$\text{Xi}_{N,1/2}/\text{Xi}_{N,0}$
96	[0.632933594160, 0.633142026033]
128	[0.716167972521, 0.716468259088]
256	[0.600909229748, 0.601137218046]

and hence all lie strictly between 1/2 and 3/4.

Proof. The finite operator and projection are the frozen TPC-269 objects. The interval engine first certifies positive R^2 , applies (2), and then applies (3) to the listed pairs. Every displayed classification is separated from its stated threshold. The independent checker recomputes the source, operator, projection, and normalization using a separate floating-point implementation; the stress audit checks the registry and all threshold inequalities. $\square$

The five adjacent base ratios give the supplementary pattern DROP, DROP, RISE, DROP, RISE, with the largest finite rise at $128 \rightarrow 192$ (above 58). The profile controls are deliberately reported as ratios at the same scale: they quantify a finite profile effect without being mistaken for a cross-scale theorem.

5 Interpretation and limits

Equation (1) is useful because it turns the radius question into a dimensionless positive quantity and makes cross-scale comparisons auditable. The main finite obstruction is not the absolute size of any one row; it is the coexistence of a large normalized rise and a strict normalized drop under one declared cutoff rule and one base profile family.

Four promotions are prohibited. First, the six scales do not establish an asymptotic sequence or a uniform source-level estimate. Second, a finite rise of $D(a, b)$ does not refute an eventual bound $R_N \ll N^{5/3-\delta}$, since its constant and onset are not fixed here. Third, a finite drop does not prove such a saving. Finally, the radius product is not an arithmetic L^2 estimate and does not reassemble the signed prime-shell packet in Gate B.

6 Conclusion

TPC-270 adds the missing finite scale lens to the TPC-267–269 chain. The exact sixth-power normalization and positive interval ratios expose strong finite cross-scale variation, while matched profile controls remain in a separated subunit band. The next source-level task is to state and prove an explicit uniform radius upper bound with a declared power and clock. The actual V59 radius and phase, strict 1/400 payment, arithmetic L^2 , full Gate B, and the twin-prime conclusion remain open.

References

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